

Juneha Hwang

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🐙 <https://github.com/juneharold>

Education

Korea Advanced Institute of Science and Technology (KAIST)

August 2023-

Bachelor of Science in Computer Science (GPA: 4.01 / 4.30)

Daejeon, South Korea

- **Some relevant courses:** Introduction to Computer Networks, Machine Learning, Graph Machine Learning and Mining, Systems Programming, Computer Organization, Programming Languages

Experience

Linq Alpha

Jan 2025–Feb 2026

Data Engineer

Seoul, South Korea

- Engineered an entity resolution pipeline that maps ambiguous financial entity names to stock identifiers using a multi-stage agentic architecture (cache → trigram DB search → vector DB retrieval → LLM-based decision), combining RAG and structured reasoning for reliable production use.
- Built a semantic document chunking system that transforms unstructured financial documents (earnings calls, broker research, investment notes) into structured, queryable metadata using a multi-batch LLM pipeline—including multi-layer JSON parsing and validation to handle non-deterministic model outputs at scale.
- Designed and deployed a FastAPI-based API layer with asynchronous processing (Celery/Redis) for long-running LLM and data jobs, with observability instrumentation to monitor agent reliability in production.
- Managed large-scale financial datasets using PostgreSQL and Alembic, providing the retrieval backend for downstream AI pipelines.

Projects

DEX-DEX Perpetual Futures Arbitrage Bot | Rust, Tokio, WebSockets

- Built a low-latency arbitrage bot in Rust that detects and executes spread opportunities between two perpetual futures exchanges across 96 asset pairs in real time.
- Designed a dual-channel architecture using lock-free crossbeam channels for the hot path and Tokio broadcast for cold-path recording, with a dedicated OS thread for the signal engine to avoid async runtime contention.
- Optimized for cache performance with 64-byte aligned price state structs and atomic flags for position tracking, eliminating mutex contention on the critical path.

Papers

FinAgentBench: A Benchmark Dataset for Agentic Retrieval in Financial Question Answering

- C. Choi, J. Kwon, A. Lopez-Lira, C. Kim, M. Kim, J. Hwang, et al.
- Published at **ACM ICAIF 2025**, Singapore, Singapore

Structuring the Unstructured: A Multi-Agent System for Extracting and Querying Financial KPIs and Guidance

- C. Choi, A. Lopez-Lira, Y. Lee, J. Kwon, M. Kim, J. Hwang, et al.
- Published at **FinIR Workshop @ ACM SIGIR 2025**, Padua, Italy

Honors

Competitive Programming

- Samsung Collegiate Programming Cup (SCPC) 2024, 2025 Finalist
- International Collegiate Programming Contest (ICPC) Seoul Regional 2025 Finalist, Taichung Regional 2025 Finalist

Other Honors

- Jane Street ETC 2024 1st Place
- Fall 2023 Dean's List (KAIST)

Technical Skills

Programming Languages: Python, C / C++, Rust, Scala, JavaScript, TypeScript, Ruby, SQL, HTML/CSS

Tech Stack: PyTorch, TensorFlow, LangChain, LangGraph, Scikit-Learn, NumPy, Pandas, FastAPI, PostgreSQL, DuckDB, Celery, Redis, Docker, Kubernetes, AWS, React, Git

Languages: Korean (native), English (native), Japanese (conversational)